

1 Source Code

```
1 // ----- Shared 7447 Inputs -----
2 const int PIN_A = 2;
3 const int PIN_B = 3;
4 const int PIN_C = 4;
5 const int PIN_D = 5;
6
7 // ----- Enable pins for multiplexing -----
8 // sec1, sec10, min1, min10, hr1, hr10
9 const int EN[] = {6, 7, 8, 9, 10, 11};
10
11 // ----- Buttons (Moved to Analog pins A0-A3) -----
12 const int PAUSE_BTN = A0;
13 const int NEXT_BTN = A1;
14 const int INC_BTN = A2;
15 const int DEC_BTN = A3;
16
17 bool paused = false;
18 int lastPauseState = HIGH;
19 int lastNextState = HIGH;
20 int lastIncState = HIGH;
21 int lastDecState = HIGH;
22
23 // Digit selection when paused: 0..5 -> sec1..hr10
24 int selectedDigit = 0;
25 unsigned long lastBlink = 0;
26 bool blinkOn = true;
27
28 // ----- Current States -----
29 // Seconds
30 int W1=0, X1=0, Y1=0, Z1=1;
31 int W2=1, X2=0, Y2=1;
32 // Minutes
33 int W3=1, X3=0, Y3=0, Z3=1;
34 int W4=1, X4=0, Y4=1;
35 // Hours
36 int W5=1, X5=1, Y5=0, Z5=0;
37 int W6=0, X6=1, Y6=0;
38
39 // Timer
40 unsigned long lastSecUpdate = 0;
41
42 // Function prototype
43 void showDigit(int A, int B, int C, int D, int ENpin);
44
45 // ----- Setup -----
46 void setup() {
47     pinMode(PIN_A, OUTPUT);
48     pinMode(PIN_B, OUTPUT);
49     pinMode(PIN_C, OUTPUT);
50     pinMode(PIN_D, OUTPUT);
51
52     for (int i = 0; i < 6; i++) {
53         pinMode(EN[i], OUTPUT);
54         digitalWrite(EN[i], LOW);
55     }
56
57     pinMode(PAUSE_BTN, INPUT_PULLUP);
58     pinMode(NEXT_BTN, INPUT_PULLUP);
59     pinMode(INC_BTN, INPUT_PULLUP);
60     pinMode(DEC_BTN, INPUT_PULLUP);
61 }
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62
63 // ----- Increment Helper -----
64 void incrementDigit(int d) {
65     int A, B, C, D;
66     switch (d) {
67         case 0: { // sec ones 0--9
68             A = !W1;
69             B = (W1 && !X1 && !Z1) || (!W1 && X1);
70             C = (!X1 && Y1) || (!W1 && Y1) || (W1 && X1 && !Y1);
71             D = (!W1 && Z1) || (W1 && X1 && Y1);
72             W1 = A; X1 = B; Y1 = C; Z1 = D;
73         } break;
74         case 1: { // sec tens 0--5
75             A = !W2;
76             B = (W2 && !X2 && !Y2) || (!W2 && X2);
77             C = (W2 && X2) || (!W2 && !X2 && Y2);
78             W2 = A; X2 = B; Y2 = C;
79         } break;
80         case 2: { // min ones 0--9
81             A = !W3;
82             B = (W3 && !X3 && !Z3) || (!W3 && X3);
83             C = (!X3 && Y3) || (!W3 && Y3) || (W3 && X3 && !Y3);
84             D = (!W3 && Z3) || (W3 && X3 && Y3);
85             W3 = A; X3 = B; Y3 = C; Z3 = D;
86         } break;
87         case 3: { // min tens 0--5
88             A = !W4;
89             B = (W4 && !X4 && !Y4) || (!W4 && X4);
90             C = (W4 && X4) || (!W4 && !X4 && Y4);
91             W4 = A; X4 = B; Y4 = C;
92         } break;
93         case 4: { // hr ones
94             if (X6 == 0) { // hr tens = 0 or 1 -> 0--9
95                 A = !W5;
96                 B = (W5 && !X5 && !Z5) || (!W5 && X5);
97                 C = (!X5 && Y5) || (!W5 && Y5) || (W5 && X5 && !Y5);
98                 D = (!W5 && Z5) || (W5 && X5 && Y5);
99                 W5 = A; X5 = B; Y5 = C; Z5 = D;
100             } else { // hr tens = 2 -> 0--3
101                 A = !W5;
102                 B = (W5 && !X5) || (!W5 && X5);
103                 W5 = A; X5 = B; Y5 = 0; Z5 = 0;
104             }
105         } break;
106         case 5: { // hr tens 0--2
107             if (!(X6 == 0 && W6 == 1 && Y5 == 1)) {
108                 A = !W6 && !X6;
109                 B = W6 && !X6;
110                 W6 = A; X6 = B; Y6 = 0;
111             }
112         } break;
113     }
114 }
115
116 // ----- Decrement Helper -----
117 void decrementDigit(int d) {
118     int A, B, C, D;
119     switch (d) {
120         case 0: { // sec ones 0--9
121             A = !W1;
122             B = (!X1 && !W1 && ((!Z1 && Y1) || (Z1 && !Y1))) || (!Z1 && W1 && X1);
123             C = (!Z1 && Y1 && (X1 || W1)) || (Z1 && !X1 && !W1 && !Y1);
124             D = !X1 && !Y1 && ((Z1 && W1) || (!Z1 && !W1));

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125     W1 = A; X1 = B; Y1 = C; Z1 = D;
126 } break;
127 case 1: { // sec tens 0--5
128     A = !W2;
129     B = (Y2 && !X2 && !W2) || (!Y2 && X2 && W2);
130     C = !X2 && ((Y2 && W2) || (!Y2 && !W2));
131     W2 = A; X2 = B; Y2 = C;
132 } break;
133 case 2: { // min ones 0--9
134     A = !W3;
135     B = (!X3 && !W3 && ((!Z3 && Y3) || (Z3 && !Y3))) || (!Z3 && W3 && X3);
136     C = (!Z3 && Y3 && (X3 || W3)) || (Z3 && !X3 && !W3 && !Y3);
137     D = !X3 && !Y3 && ((Z3 && W3) || (!Z3 && !W3));
138     W3 = A; X3 = B; Y3 = C; Z3 = D;
139 } break;
140 case 3: { // min tens 0--5
141     A = !W4;
142     B = (Y4 && !X4 && !W4) || (!Y4 && X4 && W4);
143     C = !X4 && ((Y4 && W4) || (!Y4 && !W4));
144     W4 = A; X4 = B; Y4 = C;
145 } break;
146 case 4: { // hr ones
147     if (X6 == 0) {
148         A = !W5;
149         B = (!X5 && !W5 && ((!Z5 && Y5) || (Z5 && !Y5))) || (!Z5 && W5 && X5);
150         C = (!Z5 && Y5 && (X5 || W5)) || (Z5 && !X5 && !W5 && !Y5);
151         D = !X5 && !Y5 && ((Z5 && W5) || (!Z5 && !W5));
152         W5 = A; X5 = B; Y5 = C; Z5 = D;
153     } else {
154         A = !W5;
155         B = (X5 && W5) || (!X5 && !W5);
156         W5 = A; X5 = B; Y5 = 0; Z5 = 0;
157     }
158 } break;
159 case 5: { // hr tens 0--2
160     if (!(X6 == 0 && W6 == 0 && Y5 == 1)) {
161         A = X6 && !W6;
162         B = !X6 && !W6;
163         W6 = A; X6 = B; Y6 = 0;
164     }
165 } break;
166 }
167 }
168
169 // ----- Main Loop -----
170 void loop() {
171     // Pause toggle
172     int pauseState = digitalRead(PAUSE_BTN);
173     if (pauseState == LOW && lastPauseState == HIGH) {
174         paused = !paused;
175         if (paused) selectedDigit = 0;
176     }
177     lastPauseState = pauseState;
178
179     // Next cursor
180     int nextState = digitalRead(NEXT_BTN);
181     if (paused && nextState == LOW && lastNextState == HIGH) {
182         selectedDigit = (selectedDigit + 1) % 6;
183     }
184     lastNextState = nextState;
185
186     // Increment
187     int incState = digitalRead(INC_BTN);

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188     if (paused && incState == LOW && lastIncState == HIGH) {
189         incrementDigit(selectedDigit);
190     }
191     lastIncState = incState;
192
193     // Decrement
194     int decState = digitalRead(DEC_BTN);
195     if (paused && decState == LOW && lastDecState == HIGH) {
196         decrementDigit(selectedDigit);
197     }
198     lastDecState = decState;
199
200     // Blink timer
201     if (paused && millis() - lastBlink >= 500) {
202         blinkOn = !blinkOn;
203         lastBlink = millis();
204     }
205
206     // Display Multiplexing
207     for (int d = 0; d < 6; d++) {
208         bool showThis = true;
209         if (paused && d == selectedDigit && !blinkOn) showThis = false;
210
211         if (showThis) {
212             switch (d) {
213                 case 0: showDigit(W1, X1, Y1, Z1, EN[0]); break;
214                 case 1: showDigit(W2, X2, Y2, 0, EN[1]); break;
215                 case 2: showDigit(W3, X3, Y3, Z3, EN[2]); break;
216                 case 3: showDigit(W4, X4, Y4, 0, EN[3]); break;
217                 case 4: showDigit(W5, X5, Y5, Z5, EN[4]); break;
218                 case 5: showDigit(W6, X6, Y6, 0, EN[5]); break;
219             }
220             delay(2); // Short display dwell time
221         } else {
222             digitalWrite(EN[d], LOW);
223         }
224     }
225
226     // Time counting update
227     if (!paused && millis() - lastSecUpdate >= 1000) {
228         lastSecUpdate = millis();
229
230         // Seconds Ones
231         int A = !W1;
232         int B = (W1 && !X1 && !Z1) || (!W1 && X1);
233         int C = (!X1 && Y1) || (!W1 && Y1) || (W1 && X1 && !Y1);
234         int D = (!W1 && Z1) || (W1 && X1 && Y1);
235         W1 = A; X1 = B; Y1 = C; Z1 = D;
236
237         // Seconds Tens
238         if ((W1 | X1 | Y1 | Z1) == 0) {
239             A = !W2;
240             B = (W2 && !X2 && !Y2) || (!W2 && X2);
241             C = (W2 && X2) || (!W2 && !X2 && Y2);
242             W2 = A; X2 = B; Y2 = C;
243         }
244
245         // Minutes Ones
246         if ((W1 | X1 | Y1 | Z1 | W2 | X2 | Y2) == 0) {
247             A = !W3;
248             B = (W3 && !X3 && !Z3) || (!W3 && X3);
249             C = (!X3 && Y3) || (!W3 && Y3) || (W3 && X3 && !Y3);
250             D = (!W3 && Z3) || (W3 && X3 && Y3);

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251     W3 = A; X3 = B; Y3 = C; Z3 = D;
252
253     // Minutes Tens
254     if ((W3 | X3 | Y3 | Z3) == 0) {
255         A = !W4;
256         B = (W4 && !X4 && !Y4) || (!W4 && X4);
257         C = (W4 && X4) || (!W4 && !X4 && Y4);
258         W4 = A; X4 = B; Y4 = C;
259     }
260
261     // Hours Ones
262     if ((W3 | X3 | Y3 | Z3 | W4 | X4 | Y4) == 0) {
263         if (X6 == 0) {
264             A = !W5;
265             B = (W5 && !X5 && !Z5) || (!W5 && X5);
266             C = (!X5 && Y5) || (!W5 && Y5) || (W5 && X5 && !Y5);
267             D = (!W5 && Z5) || (W5 && X5 && Y5);
268             W5 = A; X5 = B; Y5 = C; Z5 = D;
269         } else {
270             A = !W5;
271             B = (W5 && !X5) || (!W5 && X5);
272             W5 = A; X5 = B; Y5 = 0; Z5 = 0;
273         }
274
275         // Hours Tens
276         if ((W5 | X5 | Y5 | Z5) == 0) {
277             A = !W6 && !X6;
278             B = W6 && !X6;
279             W6 = A; X6 = B; Y6 = 0;
280         }
281     }
282 }
283 }
284 }
285
286 // ----- Display Helper -----
287 void showDigit(int A, int B, int C, int D, int ENpin) {
288     for (int i = 0; i < 6; i++) {
289         digitalWrite(EN[i], LOW);
290     }
291
292     digitalWrite(PIN_A, A);
293     digitalWrite(PIN_B, B);
294     digitalWrite(PIN_C, C);
295     digitalWrite(PIN_D, D);
296
297     digitalWrite(ENpin, HIGH);
298 }

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